



Barker College

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Student Number

Mathematics

2015 TRIAL HIGHER SCHOOL CERTIFICATE

AM Friday 31 July

Section 1 – Multiple Choice

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐ (B) ☐ (C) ☐ (D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

correct
↓

Start
Here



1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

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Barker College

Mathematics

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Student Number

2015 TRIAL HIGHER SCHOOL CERTIFICATE

Staff Involved:

- AJD* • PJR*
- LMD • DZP
- VAB • GPF
- ARM • WMD
- GIC • JGD

AM Friday 31 July

120 copies

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Write your Barker Student Number on all pages of your solutions
- In Questions 11 – 16, show all relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2 - 3

10 marks

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II Pages 5 - 12

90 marks

- Attempt Questions 11 - 16
- Allow about 2 hours and 45 minutes for this section

Section 1 - Multiple Choice (10 marks)

Attempt Questions 1 - 10

Use the multiple-choice answer sheet for Questions 1 - 10

1. $\log_4 2 =$

- (A) $\frac{1}{2}$ (B) 1 (C) 2 (D) 4

2. Solve $|3 - x| \geq 6$

- (A) $x \geq 3$ or $x \leq 9$ (B) $x \leq -3$ or $x \geq 9$
(C) $x \leq -3$ or $x \leq 9$ (D) $-3 \leq x \leq 9$

3. The limiting sum of the series $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \dots$ is

- (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $1\frac{1}{2}$ (D) 2

4. If $5\sqrt{2} - \sqrt{8} + \sqrt{32} = \sqrt{x}$, the value of x is:

- (A) 26 (B) $\sqrt{98}$ (C) 98 (D) 130

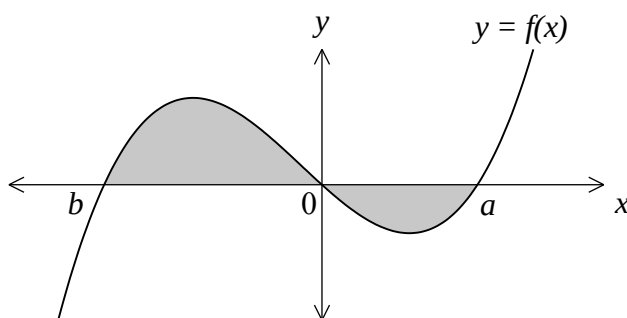
5. The derivative of $\cos 2x$ is:

- (A) $-2\cos 2x$ (B) $2\cos 2x$ (C) $2\sin 2x$ (D) $-2\sin 2x$

6. Differentiate $\frac{x\sqrt{x}}{x^5}$

- (A) $\frac{-7}{2x^4\sqrt{x}}$ (B) $\frac{2\sqrt{x}}{7x^3}$ (C) $\frac{7}{2x^2\sqrt{x}}$ (D) $\frac{-7}{2x^3}$

7. Two-digit numbers are formed from the digits 2, 3, 4, 6 with no repetition of digits allowed. A two-digit number is then selected at random. What is the probability that the number is prime?
- (A) $\frac{1}{12}$ (B) $\frac{1}{8}$ (C) $\frac{1}{6}$ (D) $\frac{5}{12}$
8. The solution of the inequality $x^2 - 6x + 8 \leq 0$ is:
- (A) $x \geq 4$ or $x \leq 2$ (B) $2 \leq x \leq 4$
 (C) $2 < x < 4$ (D) $x \leq 4$ or $x \geq 2$
9. The quadratic equation with roots k and $3k$ is:
- (A) $(x-k)(x-3) = 0$ (B) $x^2 - 4kx + 3k^2 = 0$
 (C) $x^2 + 4kx + 3k^2 = 0$ (D) $x^2 + 3k^2x - 4k = 0$
10. Which of the following correctly finds the shaded area in this diagram?



- (A) $\int_b^a f(x) dx$ (B) $\left| \int_b^a f(x) dx \right|$
 (C) $\left| \int_0^a f(x) dx \right| + \int_b^0 f(x) dx$ (D) $\int_0^a f(x) dx + \left| \int_b^0 f(x) dx \right|$

End of Section I

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Section II – Extended Response (90 marks)

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question on a separate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)	[START A NEW BOOKLET]	Marks
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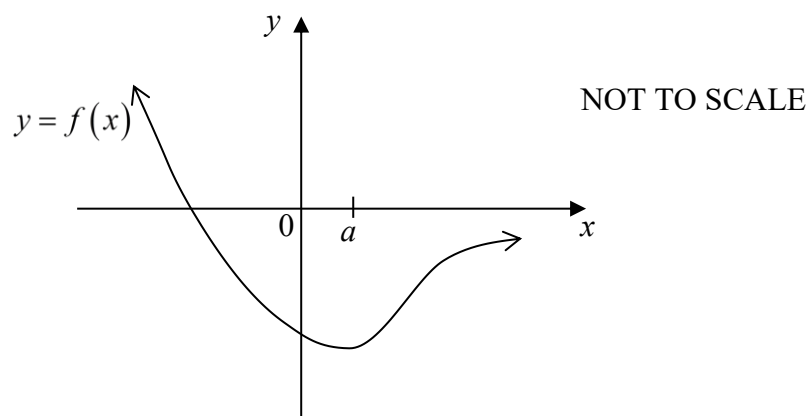
(a) Factorise fully: $4x^3 - 32$		2
----------------------------------	--	---

(b) Find: $\lim_{x \rightarrow 0} \frac{x^2 - 4x}{2x}$		1
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(c) Find the integers a and b such that: $\frac{4\sqrt{3}}{\sqrt{7} + \sqrt{3}} = a + b\sqrt{21}$		2
---	--	---

(d) Copy the following graph of $y = f(x)$ into your answer booklet.		2
--	--	---

Hence, clearly sketch and label $y = f'(x)$



Question 11 continues on page 6

Question 11 (continued)

(e) Solve for x : $4^x - 5 \times 2^x + 4 = 0$ **2**

(f) Find the domain of the function $f(x) = \sqrt{x^2 + x - 6}$ **2**

(g) (i) Use Simpson's Rule with three function values to find an approximation to the area under the curve $y = \frac{1}{x}$ between $x = m$ and $x = 3m$, where $m > 0$ **2**

(ii) Hence, using integration, show that $\log_e 3 \approx \frac{10}{9}$ **2**

End of Question 11

Question 12 (15 marks)

[START A NEW BOOKLET]

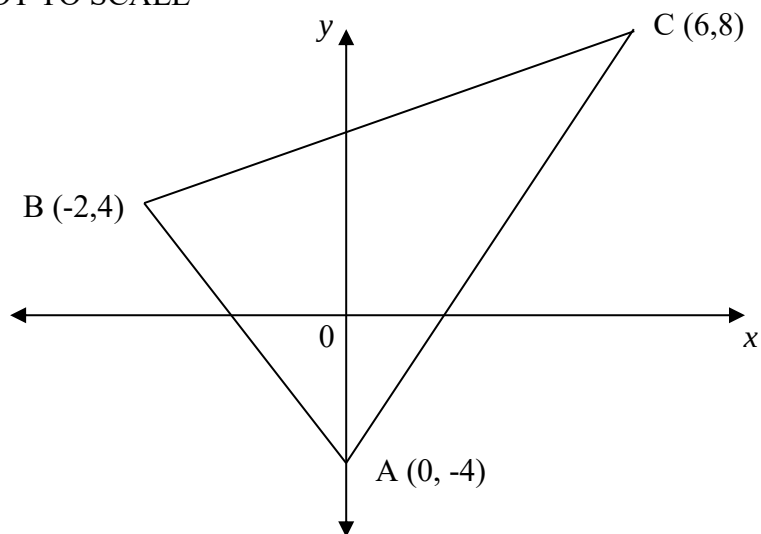
Marks

- (a) Evaluate the arithmetic series $180 + 165 + 150 + \dots -15 -30$ **2**
- (b) (i) Sketch the curve $y = \frac{3}{x-1}$ showing any intercepts. **2**
- (ii) State the domain and range of $y = \frac{3}{x-1}$ **2**
- (iii) Hence, evaluate $\int_2^4 \frac{3}{x-1} dx$ **2**
- (c) Find
- (i) $\int 3e^{7x} dx$ **1**
- (ii) $\int \tan^2 x dx$ **2**
- (d) The roots of the equation $x^2 + 4x + 1 = 0$ are α and β . Find:
- (i) $\alpha + \beta$ **1**
- (ii) $\alpha\beta$ **1**
- (iii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ **2**

End of Question 12

(a)

NOT TO SCALE



In the diagram the coordinates of the points are A(0, -4), B(-2, 4) and C(6, 8).

- | | | |
|-------|---|---|
| (i) | Show the equation of line BC is $x - 2y + 10 = 0$ | 2 |
| (ii) | Find the perpendicular distance from A to the line BC | 2 |
| (iii) | Find point D such that ABCD is a parallelogram. | 1 |
| (iv) | Hence, find the area of the parallelogram ABCD | 2 |

Question 13 continues on page 9

Question 13 (continued)

(b) Differentiate with respect to x .

(i) $y = \sqrt{7-3x^2}$ **2**

(ii) $y = \frac{3x}{\ln x}$ **2**

(c) The equation of the tangent to the curve $f(x) = 2e^x(x^2+1)$ at the point $(0, 2)$ **4**

is given by $y = ax + b$

Evaluate a and b .

End of Question 13

- (a) Sketch the parabola

$$(y + 2)^2 = -4(x + 3) \text{ clearly indicating the focus and directrix.}$$

2

- (b) A deck contains six cards labelled the numbers 0, 1, 2, 2, 3 and 3 respectively.
Ann draws two cards at random without replacement.

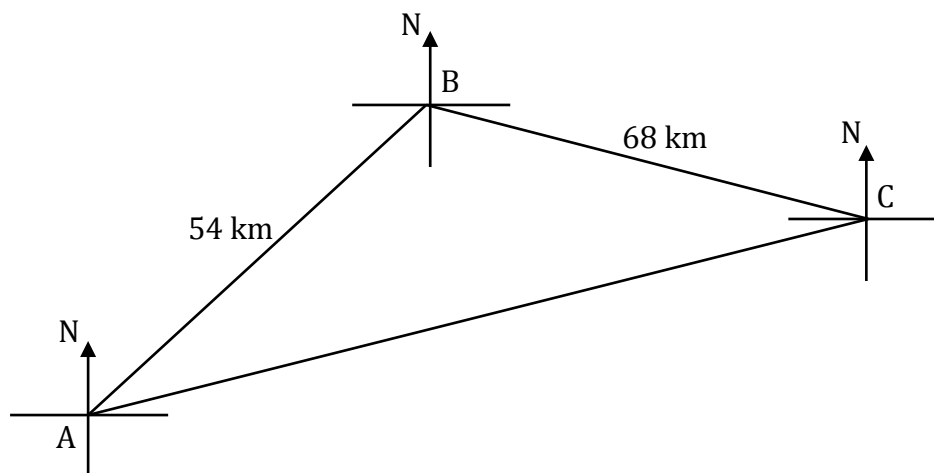
- (i) Find the probability that the sum of the two cards equals 5

1

- (ii) Find the probability that the sum of the two cards is less than five

2

- (c)



A plane flies 54 km from A to B on a bearing of 055° .

The plane then continues onto C, flying a distance of 68 km on a bearing of 105° .

- (i) Copy the diagram into your answer booklet and use this diagram to show why $\angle ABC = 130^\circ$

1

- (ii) Find the distance CA (to nearest 0.1 km).

2

- (iii) Hence, calculate the size of $\angle BAC$ (to nearest degree).

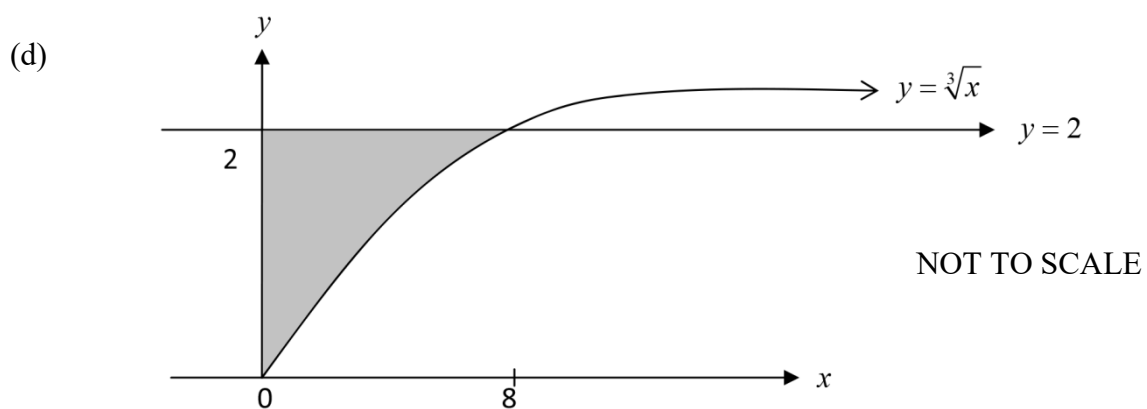
2

- (iv) Hence, or otherwise, find the bearing of A from C (to nearest degree).

1

Question 14 continues on page 11

Question 14 (continued)



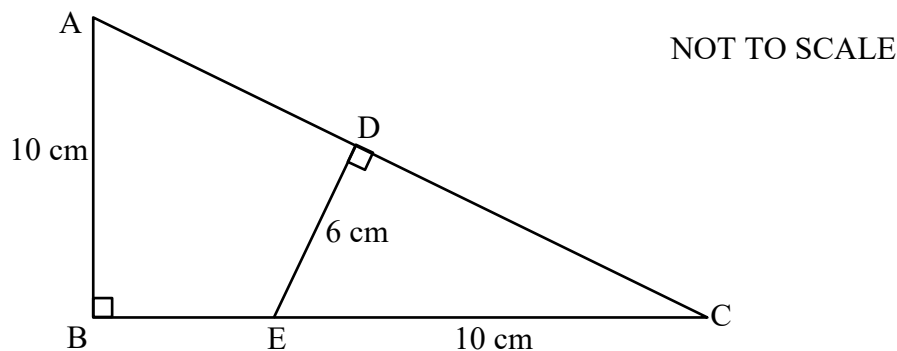
The above diagram shows the region bounded by the curve $y = \sqrt[3]{x}$, the y axis and the line $y = 2$.

Given that the point of intersection of $y = \sqrt[3]{x}$ and $y = 2$ is $(8, 2)$, find the exact volume of the solid formed when the region shown is rotated about the x -axis.

4

End of Question 14

- (a) Triangles ABC and CDE are right angled at B and D respectively as shown in the diagram.



- (i) Prove that $\triangle ABC \parallel \triangle CDE$ 2
- (ii) If $AB = EC = 10$ cm and $DE = 6$ cm, determine the length of AC . 2
- (b) Over the interval $a \leq x \leq b$, a curve $y = f(x)$ has the following properties: 2

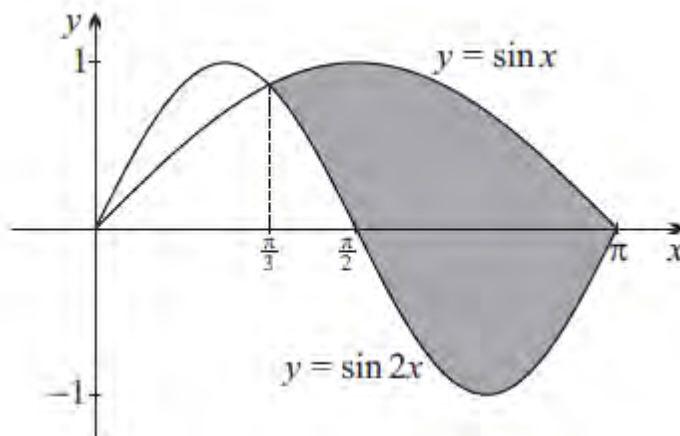
$$f(a) < 0, \quad f'(x) > 0 \quad \text{and} \quad f''(x) < 0$$

Draw this section of the curve $y = f(x)$ illustrating all of the above information.

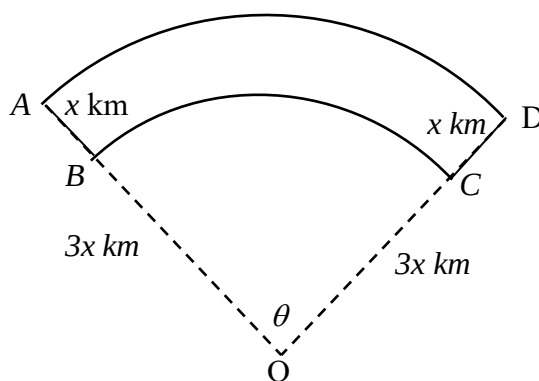
- (c) Starting with the number 3, Mike multiplies this by a whole number n . 3
 After writing down the result, he then multiplies this answer by n again and continues multiplying until he reaches 3072. In total, he multiplies by n ten times.
 He then adds up all his answers (including the original 3).
 Find the total Mike calculated.

Question 15 continues on page 13

- (d) The diagram below shows the curves $y = \sin 2x$ and $y = \sin x$ for $0 \leq x \leq \pi$ which intersect at the values $x = 0$, $x = \frac{\pi}{3}$ and $x = \pi$. Find the exact area of the shaded region bounded by these two curves. 3



- (e) Four towns A , B , C and D are joined by roads that are either straight or arcs of concentric circles with centre at O . Town B and C are $3x$ km from O . Towns A and D are both x km from B and C respectively. $\angle AOD = \theta$ radians as shown in the diagram below.



- (i) Write an expression in terms of x and θ for the length of arc AD 1
- (ii) A salesperson wants to travel from Town A to Town D but must visit town B and C on the way. Write an expression in terms of θ and x , for the length of this journey from Town A to Town D . 1
- (iii) Find the value of θ for which the journeys in (i) and (ii) are the same distance. 1

End of Question 15

- (a) The curve $y = f(x)$ has two stationary points at $x = 1$ and $x = a$ 2

If $f''(x) = 6x - 2$, find the value of a .

- (b) Solve the equation: 3

$$2\cos^2 x = 2\sin x \cos x \quad \text{where } 0 \leq x \leq 2\pi$$

- (c) Rachael borrows $\$P$ for an overseas holiday.
This loan plus interest is to be repaid in equal monthly instalments of $\$232$ over five years.
Interest of 6% p.a. is charged monthly on the balance owing at the start of each month.

Let A_n be the amount owing at the end of n months.

(i) Show that $A_2 = P(1.005)^2 - 232(1 + 1.005)$ 2

(ii) Prove that $A_n = P(1.005)^n - 46\,400(1.005^n - 1)$ 2

(iii) Hence, or otherwise, find the amount that Rachael borrowed. 2

- (d) A basketball coach knows with certainty that the number of points scored during a season by a player with shirt number n is given by: 4

$$P = n^3 - 30n^2 + 225n$$

If the shirt numbers are only whole numbers from 1 to 20 inclusive, which player(s) will score the most points in this season and determine their score(s)?

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q1 a) $\log_4 2 = x$

$$4^x = 2$$

$$2^{2x} = 2^1$$

$$2x = 1 \quad x = \frac{1}{2} \quad A$$

2) $|3-x| > 6$

$$3-x > 6 \quad -3+x > 6$$

$$-x > 3 \quad x > 9 \quad B$$

$$x < -3$$

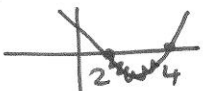
3) $a=1 \quad S_{\infty} = \frac{1}{1-\frac{1}{2}} = \frac{1}{\frac{1}{2}} = 2$
 $r = -\frac{1}{2} \quad B$

4) $5\sqrt{2} - 2\sqrt{2} + 4\sqrt{2} = 7\sqrt{2} = \sqrt{98}$
 $\therefore \sqrt{x} = \sqrt{98}$
 $x = 98 \quad C$

5) $\frac{d}{dx} \cos 2x = -2 \sin 2x \quad D$

6) $\frac{x' \times x^{\frac{1}{2}}}{x^5} = x^{-\frac{7}{2}}$
 $\frac{dy}{dx} = -\frac{7}{2} x^{-\frac{9}{2}} = \frac{-7}{2\sqrt{x^9}}$
 $= \frac{-7}{2x^4\sqrt{x}} \quad A$

7) $2 \leq 3 \checkmark$
 $3 \leq 4 \checkmark$
 $4 \leq 3 \checkmark$
 $6 \leq$
 $4 \times 3 = 12 \text{ combinations}$
 $\frac{2}{12} = \frac{1}{6} \quad C$

8) $(x-2)(x-4) \leq 0$

 $2 \leq x \leq 4 \quad B$

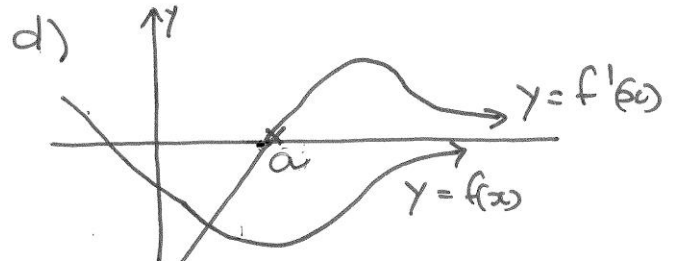
9) $(x-k)(x-3k) = 0$
 $x^2 - 4kx + 3k^2 = 0 \quad B$

10) C

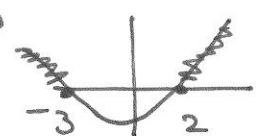
11) a) $4(x^3 - 8) = 4(x-2)(x^2+2x+4)$

b) $\lim_{x \rightarrow 0} \frac{x(x-4)}{2x} = \lim_{x \rightarrow 0} \frac{x-4}{2}$
 $= \frac{0-4}{2} = -2$

11) c) $\frac{4\sqrt{3}}{\sqrt{7}+\sqrt{3}} \times \frac{\sqrt{7}-\sqrt{3}}{\sqrt{7}-\sqrt{3}} = \frac{4\sqrt{21}-4 \times 3}{7-3}$
 $= \frac{4(\sqrt{21}-3)}{4}$
 $= \sqrt{21}-3$
 $a = -3$
 $b = 1$



e) $2^{2x} - 5 \times 2^x + 4 = 0$
 $m^2 - 5m + 4 = 0$
 $(m-1)(m-4) = 0 \quad m = 1, 4$
 $2^x = 2^0, 2^2 \quad x = 0, 2$

f) $x^2 + x - 6 \geq 0$
 $(x+3)(x-2) \geq 0$

 $x \leq -3 \text{ or } x \geq 2$

g) i)

x	m	$2m$	$3m$
y	$\frac{1}{m}$	$\frac{1}{2m}$	$\frac{1}{3m}$

$$A = \frac{m}{3} \left(\frac{1}{m} + \frac{1}{3m} + \frac{1}{2m} \right)$$

$$= \frac{m}{3} \times \frac{10}{3m} = \frac{10}{9} u^2$$

ii) $\int_m^{3m} \frac{1}{x} dx = [\ln x]_m^{3m}$
 $= \ln 3m - \ln m$
 $= \ln \left(\frac{3m}{m} \right)$
 $= \log_e 3 \approx 10/9$

$$12) a) a=180, d=-15$$

$$l = a + (n-1)d$$

$$-30 = 180 + (n-1)(-15)$$

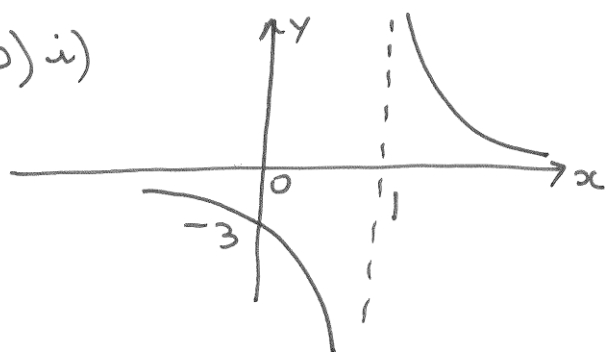
$$-210 = -15n + 15$$

$$-15n = -225$$

$$n = 15$$

$$S_{15} = \frac{15}{2}(180 - 30) = 1125$$

b) i)



$$\text{ii) } x \text{ all real } x \neq 1 \\ y \text{ all real } y \neq 0$$

$$\text{iii) } \int_2^4 \frac{3}{x-1} dx = 3 \ln(x-1) \Big|_2^4 \\ = 3(\ln 3 - \ln 1) = 3 \ln 3$$

$$c) \text{ i) } \frac{3}{7} \int 7e^{7x} dx = \frac{3}{7} e^{7x} + C$$

$$\text{ii) } \int \tan^2 x dx = \int \sec^2 x - 1 dx \\ = \tan x - x + C$$

$$d) a=1 \quad \text{ii) } \alpha + \beta = \frac{-4}{1} = -4 \\ b=4 \\ c=1 \quad \text{ii) } \alpha\beta = \frac{1}{1} = 1$$

$$\text{iii) } \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\beta^2 + \alpha^2}{\alpha^2 \beta^2} \\ = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha^2 \beta^2} \\ = \frac{16 - 2}{1} = 14$$

$$13) \text{ i) } m_{BC} = \frac{8-4}{6+2} = \frac{4}{8} = \frac{1}{2}$$

$$y - 4 = \frac{1}{2}(x + 2)$$

$$2y - 8 = x + 2$$

$$x - 2y + 10 = 0$$

$$\text{ii) } d = \frac{|1(0) - 2(-4) + 10|}{\sqrt{1^2 + 2^2}} \\ = \left| \frac{18}{\sqrt{5}} \right| = \frac{18}{\sqrt{5}} \text{ u}^2$$

$$\text{iii) } M_{AC} = (3, 2)$$

$$(-2, 4) \quad (3, 2) \quad D(x, y)$$

$$D\left(\frac{-2+x}{2}, \frac{4+y}{2}\right) = (3, 2)$$

$$\frac{-2+x}{2} = 3 \\ x = 8$$

$$\frac{4+y}{2} = 2 \\ y = 0$$

$$D(8, 0)$$

$$\text{iv) } \text{Area} = b \times ph$$

$$BC = \sqrt{8^2 + 4^2} = 4\sqrt{5} \times \frac{18}{\sqrt{5}} \\ = \sqrt{80} = 4\sqrt{5} = 72 \text{ u}^2$$

$$b) \text{ i) } y = (7 - 3x^2)^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2}(7 - 3x^2)^{-1/2} \times -6x \\ = -3x(7 - 3x^2)^{-1/2}$$

$$\text{ii) } \frac{dy}{dx} = \frac{\ln x \times 3 - 3x \times \frac{1}{x}}{(\ln x)^2} \\ = \frac{3 \ln x - 3}{(\ln x)^2}$$

$$13c) f(x) = 2e^x(x^2+1)$$

$$f'(x) = 2e^x(x^2+1) + 2e^x \times 2x$$

$$= 2e^x(x^2+2x+1)$$

$$f'(0) = 2(1) = 2$$

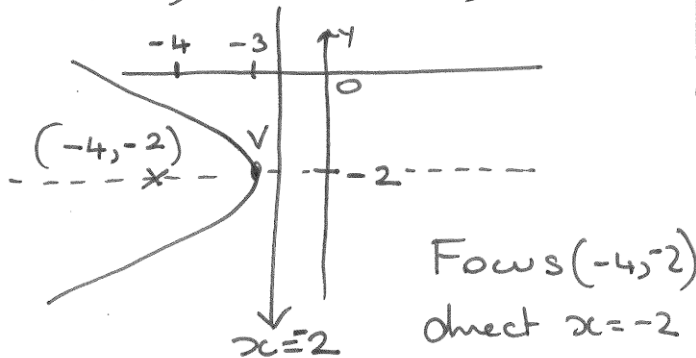
$$x=0 \quad f(0) = 2(1) = 2$$

$$y-2 = 2(x-0)$$

$$y = 2x + 2$$

$$\therefore a = 2, \quad b = 2$$

Q14 a) $V = (-3, -2) \quad a = 1$

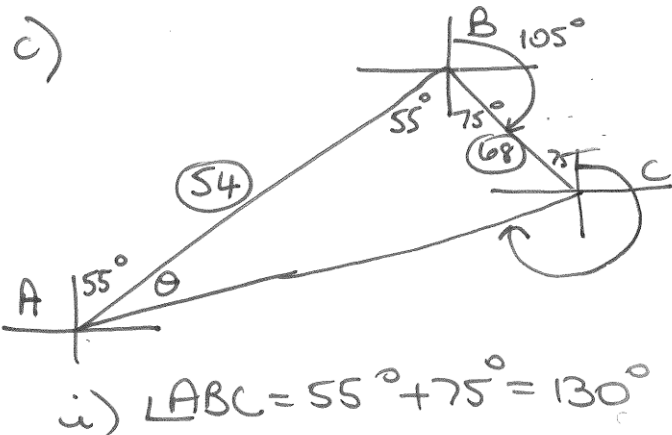


b) $i) P(\text{Sum} \leq 5) = \frac{2}{6} \times \frac{2}{5} + \frac{2}{6} \times \frac{2}{5}$

$$= \frac{4}{15}$$

$ii) P(\text{Sum} < 5) = 1 - \frac{4}{15} = \frac{11}{15}$

$$= \frac{20}{30} = \frac{2}{3}$$



$$ii) \angle ABC = 55^\circ + 75^\circ = 130^\circ$$

$$11) CA^2 = 54^2 + 68^2 - 2 \times 54 \times 68 \times \cos 130^\circ$$

$$= 12260.63 \dots$$

$$CA = 110.727 = 110.7 \text{ km}$$

$$111) \frac{\sin \theta}{68} = \frac{\sin 130^\circ}{110.7}$$

$$\sin \theta = \frac{68 \sin 130^\circ}{110.7} = 0.4705 \dots$$

$$\theta = 28^\circ$$

$$\angle C = 180 - 130 - 28 = 22^\circ$$

$$\text{Bear} = 360 - (75 + 22)$$

$$= 263^\circ$$

$$d) V = \pi \int_0^8 2^2 dx - \pi \int_0^8 x^{2/3} dx$$

$$[V = \pi \int y^2 dx]$$

$$= \pi \int_0^8 4 - x^{2/3} dx$$

$$= \pi \left[4x - \frac{3}{5} x^{5/3} \right]_0^8$$

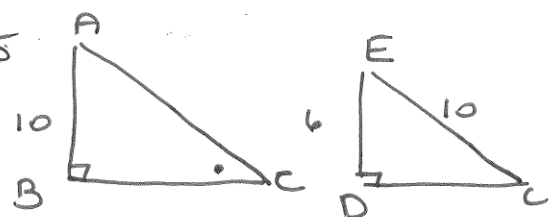
$$= \pi \left[32 - \frac{3}{5} \sqrt[3]{8^5} - 0 \right]$$

$$= \pi \left(32 - \frac{3 \times 32}{5} \right)$$

$$= \frac{64\pi}{5} \text{ u}^3$$

Q15

a) 10



In $\triangle ABC$ and $\triangle EDC$

$$\angle ABC = \angle EDC = 90^\circ \text{ (given)}$$

$$\angle ACB = \angle ECD \text{ (common } \angle)$$

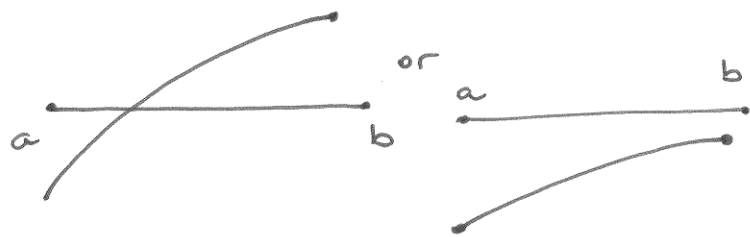
$\therefore \triangle ABC \parallel \triangle EDC$ (equiangular)

$$ii) \frac{AC}{EC} = \frac{AB}{ED} \text{ (ratio of corresp sides in similar } \Delta's)$$

$$\frac{AC}{10} = \frac{10}{6}$$

$$AC = \frac{100}{6} = 16\frac{2}{3} \text{ cm}$$

Q15b)



c) $3, 3n, 3n^2, \dots, 3n^{10} \leftarrow 11 \text{ terms}$

$3, \dots, 3072$

$3072 = 3n^{10}$

$n^{10} = 1024 \therefore n = 2$
 $= 2^{10}$

$3, 6, 12, \dots, 3072$

$S_{11} = \frac{3(2^{11}-1)}{2-1} = 6141$

d) $A = \int_{\pi/3}^{\pi} \sin x \, dx - \int_{\pi/3}^{\pi} \sin 2x \, dx$

$= -\left[\cos x\right]_{\pi/3}^{\pi} + \frac{1}{2}\left[\cos 2x\right]_{\pi/3}^{\pi}$

$= -(-1 - \frac{1}{2}) + \frac{1}{2}(1 - \frac{1}{2})$

$= 1\frac{1}{2} + \frac{3}{4}$

$= 2.25 \text{ or } 2$

e) i) $AD = 4x\theta$

ii) $ABCD = x + 3x\theta + x$

$= 2x + 3x\theta$

iii) $AD = ABCD$

$4x\theta = 2x + 3x\theta$

$x\theta = 2x$

$x(\theta - 2) = 0$

$\therefore x \neq 0 \quad \theta = 2 \text{ radians}$

Q16

a) $f''(x) = 6x - 2$

$f'(x) = 3x^2 - 2x + C$

$\frac{dy}{dx} = 0 \quad 0 = 3 - 2 + C$
 $C = -1$

$f(a) = 0 = 3a^2 - 2a - 1$

$(3a+1)(a-1) = 0$

$a = -\frac{1}{3}, 1$

$\therefore a = -\frac{1}{3}$

b) $2\cos^2 x - 2\sin x \cos x = 0$

$2\cos x (\cos x - \sin x) = 0$

$\cos x = 0$

~~$\cos x = 0$~~

$\cos x = \sin x$

$\tan x = 1 \checkmark$

$x = \pi/2, 3\pi/2$

$x = \pi/4, 5\pi/4$

$x = \pi/4, \pi/2, 5\pi/4, 3\pi/2$

c) $A_1 = P(1.005) - 232$

i) $A_2 = [P(1.005) - 232](1.005) - 232$

$= P(1.005)^2 - 232(1.005) - 232$

$= P(1.005)^2 - 232(1 + 1.005)$

ii) $A_n = P(1.005)^n - 232(1 + 1.005 + \dots + 1.005^{n-1})$

$= P(1.005)^n - \frac{232 \times 1(1.005^n - 1)}{.005}$

$= P(1.005)^n - 46400(1.005^n - 1)$

iii) $A_{60} = 0$ amt paid back

$\therefore P(1.005)^{60} - 46400(1.005^{60} - 1) = 0$

$P = \frac{46400(1.005^{60} - 1)}{1.005^{60}}$

$= \$12000.33$

$\approx \$12000$

Q.16d)

$$P = n^3 - 30n^2 + 225n$$

$$\frac{dP}{dn} = 3n^2 - 60n + 225$$

$$= 3(n^2 - 20n + 75) \quad \text{stat pts}$$

$$0 = 3(n-5)(n-15) \quad \frac{dP}{dn} = 0$$

Stat pts $n = 5, 15$

$$\frac{d^2P}{dn^2} = 6n - 60$$

$$f''(5) = 30 - 60 = -30 < 0$$

max value $n = 5$

$$f''(15) = 90 - 60 = 30 > 0$$

min value $n = 15$

test Endpoints $n = 1$ $P = 196$
 $n = 20$

$$P = 20^3 - 30(20)^2 + 225(20)$$
$$= 500$$

$$n = 5 \quad P = 5^3 - 30(5)^2 + 225(5)$$
$$= 500$$

$\therefore$ Players 5 and 20 have
highest score of 500